Supraharmonics within a Datacenter-Emission and Propagation

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***Abstract*—The reliability of the electrical and cooling systems is of utmost importance for the uninterrupted operation of the data center information technology (IT) load. The electrical distribution of the data center includes many subsystems starting with the utility and building transformers, uninterruptible power supply (UPS), power distribution units (PDUs), and power supplies ultimately powering the fans and the internal components of IT equipment. The various converters in data centers emit switching frequency residue due to PWM (pulse width modulation) techniques. The switching frequency range falls within the supraharmonic range, i.e., 2 to 150 kHz. This paper aims to show, with measurements, the different types of supraharmonic emissions measured in the data center, and the difference between their maximum and average magnitudes. The paper shows a method to identify the equipment emitting supraharmonic emission and possible disturbances caused by it. The paper traces the propagation of supraharmonic emission from the source through the transformer to the grid. Lastly, a comparison of measurements is made with the compatibility levels given by standard IEC 61000-2-2.**

***Index Terms*—Data center, Information technology, Power quality, Power supplies, Pulse Width Modulation, Supraharmonics**

I. Introduction

The use of converters, due to their flexibility in controlling the power, has increased in recent years. These converters are present in all types of electronic loads with variable power ranges and are present in many devices found in our homes and industrial facilities, e.g. PV inverters, servers, LED lamps, and more. To increase the efficiency as well as to damp the harmonics (<2 kHz), they are likely to be switched in the range of 2 kHz to 150 kHz, which will be referred to as supraharmonics in this paper [1]. Industrial facilities such as data centers house converters such as uninterruptible power supply (UPS), switched-mode power supply (SMPS), and converters used in voltage and frequency drives (VFD). The converters are often operated in series and parallel to cater to the power demand as well as to increase the availability of power to the servers [2]. Pulse width modulation (PWM) switching techniques are commonly used in these converters. The converters are one of the main source of supraharmonics (non-intentional emission) in data centers. The authors in [3]-[4] study the interferences due to supraharmonics which include additional heating, audible noise, malfunction of equipment, and false tripping of residual current devices (RCD). The authors in [5] show the presence of low-frequency oscillation, called the beat frequency which results due asynchronous operation of converters. Beat frequency can cause the voltage regulators in the server power supplies to mal-operate [6]. From the perspective of data center operators, any malfunction of equipment can lead to downtime of servers causing them economic loss [7].

The behavior of supraharmonics is quite different than harmonics, hence a distinction should be made between them [1]. The supraharmonics can be of distinct types, namely narrow band, and broadband. They can vary in frequency and magnitude with time as studies by the authors in[8]. Depending on the type of supraharmonic emission, its summation in the neutral conductor varies. This leads to an increase in the PE-N (Protective earth to neutral) voltage. Work is on-going but still there is a lack of standardization when it comes to immunity and compatibility levels for supraharmonics. The compatibility levels for voltage characteristics are defined in EN50160 [9] and IEC 61000-2-2-Amd 2 [10] for differential mode for narrowband (non-intentional) emission. Although some standards have now included them as informative annex, a thorough standardization on measurement and quantification is required [11]. The industry such as data centers thus lacks the knowledge to quantify and compare the supraharmonics within their installation. Studies about supraharmonic propagation show that they propagate locally within the installation to a high degree. Information about how they would propagate within a three-phase industrial installation is not well known. The impact of voltage unbalance, load unbalance and conductor crosstalk for supraharmonics in three phases has been studied in [12] where controlled experiments formed the basis for the results. This paper intends to show the emission one can expect to find within a data center during normal

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operation. The other objective is to show the propagation of supraharmonics upstream towards the grid. Hypothesis about the cause and effect of supraharmonic emission found in the data center is made. Lastly, the emission will be compared with some of the existing standards.

II. Description of a Data Center

A. Power Electronics in a Data Center

The power electronics in data centers are the core building blocks of the electrical supply system in a data center. A block diagram of the different supply stages in a data center is shown in Figure 1. The incoming AC power from the utility is conditioned using an uninterruptible power supply (UPS) that provides a backup in case of power failure. The output of the UPS is supplied to the input power supply unit (PSU) of the server [13], through the power distribution unit (PDU).

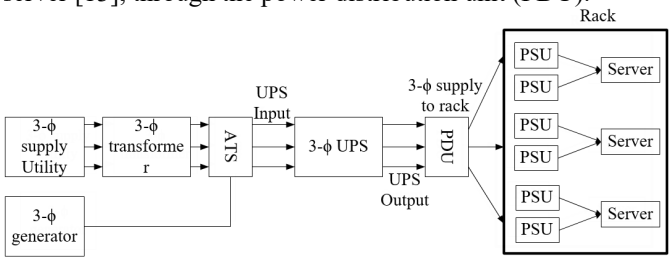


Figure 1. Block diagram representation of data center.

The server PSU further distributes the power to various applications like server fan, communication, and memory supply and point of load (PoL) converter supplying the microcontroller. Within the UPS, depending on the topology, the power undergoes conversions once or twice. The UPS is a combination of rectifier and inverter circuits. It also has a DC-DC converter for the integration of batteries into UPS, which takes care of charging and discharging them. The PSU has a power factor corrected (PFC) rectifier and a DC-DC converter circuit. A dual power supply server supply is fed by two PSUs, each of which is rated to take the full load and are supplied from separate phases. The aim is to have a backup PSU in case one PSU fails. The load is shared by the PSUs during normal operation.

B. Description of Data Center where Measurements are made

Measurements from a data center rated 24 MVA are presented in this paper. The data center is supplied by two 3-φ, 11 kV sources on the medium voltage (MV) level. The MV supply is further split into five MV feeders, three of them are supplied by source 1 and the remaining two are connected to source 2. The supply from 11 kV feeders is further stepped down to 433 V through fourteen Δ/Y transformers creating fourteen low voltage (LV) feeders. The UPSs, cooling, and lighting load is further split between these LV feeders. The data center has 5 floors, where transformers are housed on the ground floor whereas UPS rooms are housed from floors 1 to 5. Every floor has four UPS rooms, namely R1, R2, R3, and R4. Each UPS room has four UPSs operating in parallel. The LV feeders are spread to supply different UPS rooms and other loads. The measurements were conducted by using a PQube3 meter at the input panel of the transformer primary (MV), the output panel of transformer secondary (LV), and the output panel of the UPS rooms. The measuring device uses a sampling rate of 1 MHz, a measurement interval of 10 cycles, and filters per IEC 61000-4-30. The PQube3 gives the 1-minute average and maximum values of voltages with 200 Hz grouping between 2 to 9 kHz (36 groups) and 2 kHz grouping from 2 to 150 kHz at the measured locations measured according to IEC 61000-4-30 Ed. 3. The UPS output is connected to a Δ/Y transformer which further distributes power to the single-phase server loads. Due to the connection of the transformer to the output of the UPS, it is assumed that the supraharmonic emission measured at the UPS output panel has a higher contribution from the UPS as compared to the server load connected downstream.

III. Supraharmonics within the Data Center Measured at LV Terminals

Figure 2 shows the magnitude of supraharmonics measured at phase L2 of the transformer terminals of thirteen LV feeders, where the data from LV feeder 9 is not shown. The measurements are performed for a period of 12 or 24 hours at each LV feeder terminal. The 95th percentile values as suggested by IEC 61000-4-30 of the average voltage (rms) values calculated every one minute are presented in Figure 2. The magnitudes are plotted as stacked plots, where the length of the bar presents the magnitude, and the color represents the feeder number. The data center shows the presence of different frequency components chief among them in 2 to 9 kHz range are 4 kHz, 5 kHz, 5.5 kHz, and 6 kHz measured at the LV terminals. The LV feeders supply different UPSs, cooling, and lighting load, therefore, show a difference in the magnitude and the frequency of emission. The feeders are also divided into groups in terms of the MV source (11 kV) they are supplied from. Thus, the change in the upstream and downstream impedance due to the source and the load may be the cause for the change in the magnitude of emission. From the frequency spectrum shown in Figure 2, the magnitude of 4 kHz, 5 kHz, 5.5 kHz (5.4 kHz and 5.6 kHz), and 6 kHz are higher.

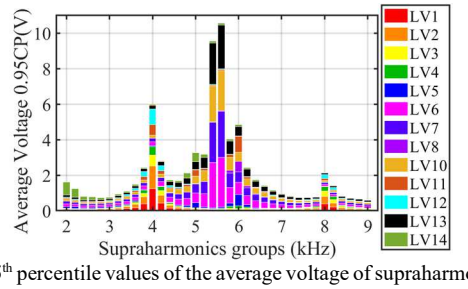


Figure 2. 95th percentile values of the average voltage of supraharmonic groups superimposed on phase L2 of different LV feeders within the data center for a period of 12 to 24 hours shown as a stacked plot.

Figure 3(a) and (b) shows a cumulative value of supraharmonic contributions (95th percentile values) from frequencies present at different LV terminals for phase L2 and phase L3 respectively. The magnitude of supraharmonic emission in phase L1 is approximately equal to that in phase L3 and hence is not shown here. It can be observed that although both the

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phases show the presence of the same frequency components, the magnitude of emissions differs.

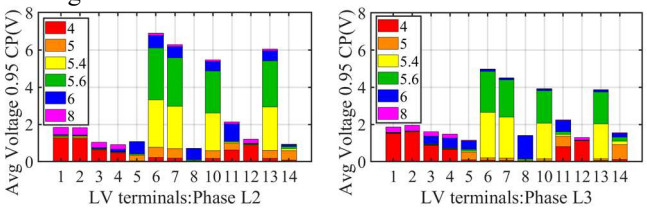


Figure 3. 95th percentile values of the average voltage of supraharmonic groups at different LV terminals for (a) phase L2 and (b) phase L3.

LV terminals 6, 7, 10, and 13 show the presence of 4 & 5 kHz components in phase L2, but they are absent or negligible in phase L3. The main emission at these terminals is 5.5 kHz, leading to the conclusion that they are connected to loads that emit 5.5 kHz. The cumulative magnitude for all the contributions on LV 6, is 7 V, which is considerable. Similarly, LV terminals 1 to 4 and 12, have 4 kHz and its multiple 8 kHz as primary emission. The 6 kHz emission is present in LV terminals 5, 8, and 11 but also present in LV terminals with a switching frequency of 5.5 kHz. This shows that the downstream impedance i.e., the load connected to each phase also differs. Although the connected loads between the three phases are balanced, some unbalance exist during the operation of servers. Further investigations are made to study the emissions at the UPS output terminals to identify the causes for the presence of these frequency components at the LV terminals.

A. Categorizing Supraharmonic Emissions within the Data Center

Figure 4 shows the average value of different types of supraharmonic emissions observed in the data center with the change in time. Figure 4(a) shows supraharmonics with 200 Hz grouping, where 5 kHz, shows the average voltage grouped between 5 kHz and 5.2 kHz, and 6 kHz is grouped between 6 kHz and 6.2 kHz measured every minute at phase L2 of LV 5 and LV 11 for one day. The measurements on LV 5 and LV 11 are done on the same day and both show the presence of 5 kHz and 6 kHz emission. As seen from Figure 4(a) observed supraharmonic emission are of different types, namely, constant frequency and magnitude, constant frequency but variable magnitude with respect to time. Figure 4(b) and Figure 5 shows a broadband emission between 60 and 72 kHz, with 2 kHz grouping measured at phase 2 of LV 11.

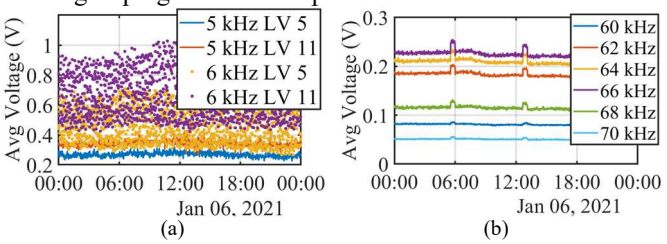


Figure 4. Change in supraharmonics emission with respect to time (a) 5 kHz (Type A) and 6 kHz (Type A1) respectively measured at LV 5 & 11 for phase L2 (b) Type B emission between 60 to 72 kHz measured at LV 11 for phase L2.

For broadband emission the magnitude remains constant with respect to time, but it is spread within a large frequency range. An increase in the emission in broadband is observed at regular intervals of seven hours. The reason for this is not known. A second broadband emission is observed between 120 to 140 kHz, which may be the multiple harmonics of 60 to 70 kHz. The emissions in the data center can thus be categorized as shown in TABLE I as defined in [8].

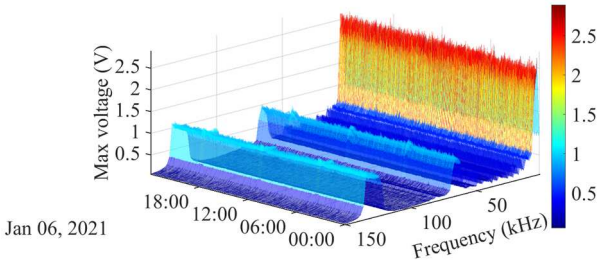


Figure 5. Change in the maximum value of 1 minute aggregated voltage at LV 11 for phase L2 for frequencies from 2 to 150 kHz, with 2 kHz grouping.

TABLE I. CATEGORISING SUPRAHARMONIC EMISSIONS WITHIN THE DATA CENTER

Type	Description	Frequency
Type A	Narrow band. Constant frequency and magnitude	4 kHz, 5 kHz And their multiples
Type A1	Narrow band. Constant frequency and variable magnitude w.r.t time	5.4 kHz, 5.6 kHz, 6 kHz And their multiples
Type B	Broadband emission	60-70 kHz, 120-140 kHz

Emission type A will show less difference in the average and maximum value per minute, whereas the emission with type A1 will show more difference between the average and maximum magnitudes. It should be noted that although the magnitude is variable for type A1, it varies within a maximum and minimum value range and does not become zero at any instant. The magnitude of the emission for the same frequency differs for LV feeders 5 and 11. The sine pulse width modulation (SPWM) and space vector pulse width modulation (SVPWM) are common PWM techniques that cause emission showing characteristics like type A. The authors hypothesize that a converter controlled to generate constant output power will generate emission of type A1. PWM employing hysteresis control is one of the PWM techniques that can generate emission of type B. Figure 5 shows the variation in the magnitude of supraharmonics from 2 to 150 kHz of phase 2 at LV 11. From Figure 4(b) and Figure 5, comparing the magnitude of broadband emission between 60 and 70 kHz, the maximum values are higher than average voltage value every minute. As seen from Figure 3(a) a frequency of 4 kHz is also present on LV 11. The multiples of 4 kHz e.g., 16 kHz, 20 kHz are seen superimposed in Figure 5. The 4 kHz component is likely due to PWM switching of equipment connected downstream of the transformer.

B. Difference between the Maximum and Average Values

Figure 6 shows the 95th percentile values of the maximum value measured at the transformer terminals on phase L2 for the LV feeders. Comparing Figure 2 and Figure 6, it can be

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said that the maximum value of the voltage can be 1.5 to 3 times the average values per supraharmonic group. The highest magnitude among different frequency groups is for 5.4 kHz and 5.6 kHz. Figure 7(a) and (b) shows the magnitude of average and maximum voltage values measured at phase L1, LV 6 for 5.4 kHz and 5.6 kHz respectively. The difference between the average and maximum values per minute varies between 5 % to 231 % w.r.t the average value. The maximum voltage value has a cyclic pattern where it varies between high and low magnitude every 15 minutes approximately. The authors hypothesize two reasons for this behaviour, one is that the supraharmonics depend on the current drawn from the device emitting them. The other is there is asynchronistic behaviour (in terms of frequency) between the multiple devices emitting them, causing a low-frequency oscillation superimposed on the supraharmonics emitted by them. In this case, the low-frequency oscillation, a so-called beat frequency, is approximately 0.06 Hz. However, further measurements are needed to confirm it.

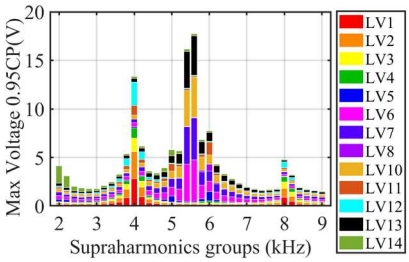


Figure 6. 95th percentile values of the maximum voltage of supraharmonics groups superimposed on phase L2 of different LV feeders within the data center for a period of 12 to 24 hours.

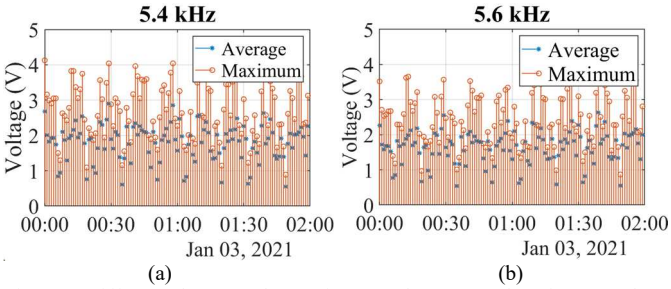


Figure 7. Difference between the maximum and average values for 5.4 and 5.6 kHz measured at LV 6 phase L1.

C. Measurements at UPS Output Terminals

Figure 8 shows the 1-minute aggregated value of maximum voltage measured at phase L1 for the UPS rooms located on the 2nd, 3rd and 4th floor of the data center measured at the output panel of respective UPS rooms. The emissions on the 1st floor matched that on the 2nd floor, and the emissions on the 5th floor are similar to those on the 4th floor and are not shown here. Similar emissions were found to be present on phase L2 and L3 albeit with different magnitudes. The output of UPS is connected to Δ/Y transformers, the secondary of which connects to the server loads. Thus, it is assumed that the UPS emission is the main emission seen at the UPS output terminal. Figure 8(a) shows that the sixteen UPS connected on the 2nd floor, are of the same type and emit 4 kHz as the main frequency, with 8 kHz as the multiple harmonic. The impedance seen by 8 kHz is low, thus its magnitude is higher than 4 kHz. The groups around 4 kHz and 8 kHz also have a higher magnitude due to the sidebands caused by PWM techniques. Figure 8(b) shows the emissions on the 3rd floor, measurements are available only for three UPS rooms.

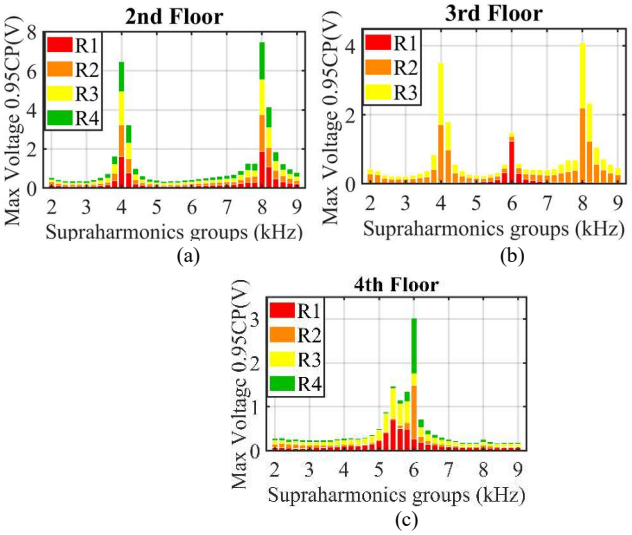


Figure 8. Stacked plot of maximum voltage emission per minute measured at phase L1 for the UPS rooms R1, R2, R3, and R4 located on (a) 2nd, (b) 3rd, and (c) 4th floor of the data center measured at the output panel of respective UPS rooms.

It can be seen that the UPSs for room R1 are of a different type. The emission from this UPS is at 6 kHz. When the emission was examined w.r.t time it was found to be of type A1. Figure 8(c) shows the emissions on the 4th floor, here also two rooms R2 and R4 house UPS with 6 kHz as the main emission whereas R1 and R3 have a switching of 5.5 kHz. All the UPSs on the 5th floor have 5.5 kHz as the switching frequency. The frequencies 4 kHz, 5.5 kHz, 6 kHz were observed on the LV terminals. It can be concluded that the most likely sources of these frequencies were the PWM switching used in the UPS.

D. Correlation Plots

To ascertain whether, for a given LV terminal, the three phases are supplying three-phase loads and single-phase loads a correlation of supraharmonic groups is used. The one-minute average values at LV terminal for one day and one phase results in a matrice of 1440×36. Correlation is calculated between supraharmonic matrices of phase L1 and L2, L2 and L3. Pearson's linear correlation coefficient is used to quantify the correlation in this study. The correlation matrix is used to investigate the correlation between different supraharmonic groups for the two phases. A high correlation coefficient between two supraharmonic groups for the same frequency, suggests that the source connected to both the phases is the same and is emitting in synchronism. A low correlation on the other hand can be interpreted as originating from different sources. Figure 9(a) and (b) show the correlation between supraharmonic matrices of phase L1 and L2, L2 and L3 for LV 11. The LV feeder 11 has three primary frequency emissions i.e., at 4 kHz, 5 kHz, and 6 kHz. A correlation factor +1 indicates the highest positive correlation and a correlation factor of -1 indicates the highest negative correlation.

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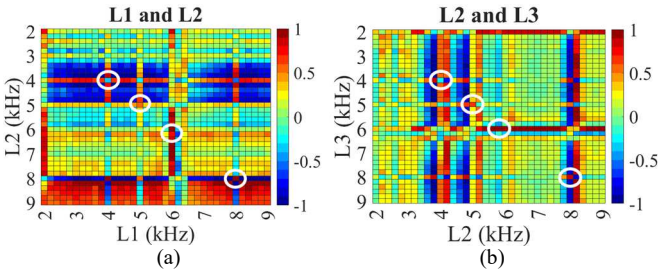


Figure 9. Correlation between supraharmonic group matrices of (a)phase L1 and L2, (b) L1 and L3 for LV 11 for emission for one day.

It can be seen from Figure 9(a) and (b) that 6 kHz has a correlation coefficient of 0.94 and 0.98 respectively suggesting that it is emitted by one specific load supplied by LV 11. The multiple of 4 kHz, i.e, 8 kHz also shows a high correlation factor. The 4 kHz emission has a negative correlation factor of -0.2 and -0.3 for both cases, which indicates that there is no correlation. The 4 kHz could be a result of conducted emission propagating from other parallel terminals connected to the same transformer. The correlation factor for 5 kHz frequency is 0.63 and 0.67 for Figure 9(a) and (b) respectively. The 5 kHz emission is not caused by UPS (c.f. Figure 8). Since there is a positive correlation it may be caused by a single-phase device connected in either of the phases or it may be a result of conductor crosstalk where a single-phase device is connected to any phase.

E. PE-N Voltage at different LV Terminals

The 1-minute aggregated value of PE-N voltage was measured at different LV terminals with 2 kHz grouping. Figure 10 exemplarily shows the 95 percentile values of PE-N voltage measured at LV terminals 2, 7, and 13 for 24 hours.

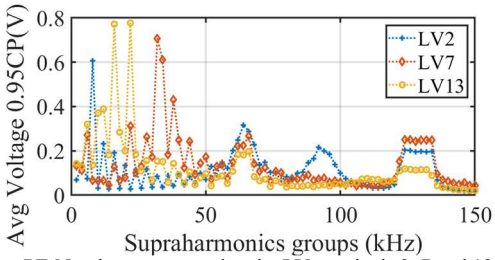


Figure 10. PE-N voltage measured at the LV terminals 2, 7 and 13 measured for 24 hours.

The terminals are chosen such that they include 4 kHz, 5.4, 5.6 kHz which are the common emissions superimposed on the voltage at different LV terminals. It can be seen from Figure 10 that all the terminals have two common broadband emission from 60 to 70 kHz and it's multiple at 120 to 140 kHz. LV 2 shows the integer multiples of 4 kHz in decreasing magnitude after 8 kHz. It also has a broadband component of 82 to 98 kHz not seen in the other feeders. LV 7 and 13 have the main emission at 5.4 and 5.6 kHz. The multiples of these frequencies are present on both terminals, but the magnitude of the individual component differs. While LV 7 has the highest amplitude at 32 kHz, LV 7 has it at 10 and 12 kHz. A common phenomenon noted at all the terminals is that in the PE-N voltage the multiples of the switching frequency emission have a higher magnitude than the switching frequency. The highest 95 percentile among the components presented in Figure 10 is close to 0.8 V, this would be superimposed on the fundamental component and add to in the overall magnitude of the PE-N voltage. The PE conductor being common between all the terminals can spread of supraharmonic frequency between LV terminals. All the emissions seen at the PE terminal can be seen at the PE-N voltage at that terminal.

IV. Propagation of Supraharmonics to MV Terminals

The supraharmonic emission seen at the output terminals of the UPS can also be seen at the LV terminals. Thus, it can be said that the supraharmonics propagate upstream. Measurements are made at the supply panel of the five MV terminals supplying the transformers within the data center. The measurements are made using potential transformers (PT) that step down the voltage on the MV side i.e., 11 kV to110 V. The PTs are old and the bandwidth of the measurement in the high-frequency range is unknown. So, the magnitude of the frequency components cannot be trusted upon. However, the presence of supraharmonic frequencies at the MV level can still be seen with measurements. Figure 11 shows the presence of the frequency components at different MV terminals.

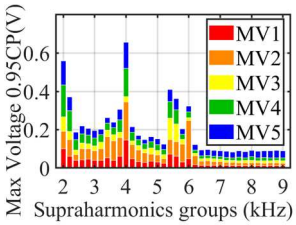


Figure 11. 95th percentile values of the maximum voltage of supraharmonics groups superimposed on phase L2 of different MV feeders within the data center for 24 hours.

The frequency emissions at 4 kHz, 5.4 kHz, 5.6 kHz, and 6 kHz seen at the UPS output terminals, which propagate to LV terminals are seen to propagate upstream towards MV terminals through the transformer. The other components that are higher in magnitude are 2 kHz and 2.2 kHz. These components could be an extension of the harmonic emission. The magnitude could be the combined contribution of harmonics propagating from upstream to the MV terminals and harmonics injected by the data center. The broadband emissions from 60 to 70 kHz and 120 to 140 kHz (not shown here) can also be seen at the MV terminals. This emission is seen to be present on other phases as well as other MV terminals.

V. Comparing the Emission with the Compatibility Limits Stated in IEC 61000-2-2

There is a gap in the compatibility and immunity standard for supraharmonics emission (non-intentional) at low voltage levels for different types of supraharmonic emissions. However, several standards cover this frequency range pertaining to intentional emission due to power line communication (PLC). The compatibility limits for voltage characteristics are defined in EN50160 and IEC 61000-2-2 for differential mode for narrowband (non-intentional) emission. The voltage characteristics according to EN 50160 is 2% of the nominal voltage for a frequency range of 2-9 kHz. IEC 61000-

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2-2-Amd 1 defines a curve with the emission level at 2 kHz equal to 1.4 % of the nominal frequency and the level decreases linearly from 1.4% to 0.65% with the logarithm of the frequency in the range 3 to 9 kHz. The measurements should be done with a bandwidth of 200 Hz. The measurements from PQube3 for the frequency range of 2 to 9 kHz are with 200 Hz bands, thus a comparison of the emissions within the datacenter is made with the compatibility limits stated in IEC 61000-2-2. For long-term measurements, the standard suggests using 10-minute aggregated values. Thus the 1-minute average values (measured according to IEC 61000-4-30) are aggregated in time over 10-minute windows and averaged according to IEC 61000-4-7 [14]. The 95th percentile value of the 10-minute average values are used for comparison with limits stated in IEC 61000-2-2. A nominal voltage of 230 V is used as a reference voltage. Figure 12 shows a comparison of the compatibility level stated in IEC 61000-2-2 with the emissions at the LV terminals for phases L1, L2, and L3.

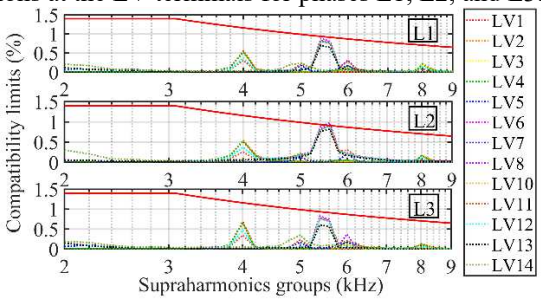


Figure 12. Comparison of compatibility level stated in IEC 61000-2-2 with the emissions at the LV terminals for phase L1, L2, and L3.

The compatibility level stated by IEC 61000-2-2 is shown by the solid line. It can be seen that the emission from 2 to 9 kHz is within limits for phases L1 and L3. The 5.4 and 5.6 kHz emission at LV 6 is above the limits. The maximum value per minute for the emissions was found to exceed the limits. However, the measurement methods stated in IEC 61000-4-30 Ed-3 used by PQube3 for 2 to 9 kHz are informative. The comparison shown in Figure 12 should therefore be treated as advisory rather than a judgement on the magnitudes at different LV terminals. Further work in the measurement and quantification of supraharmonics is needed for a standardized comparison.

VI. Conclusion

The operation of multiple converters in the data center leads to the presence of different types of supraharmonics. The magnitude of supraharmonics depends on the emitting source and the impedance of the load connected downstream or upstream and varies between phases. The supraharmonics emitted by UPS were shown to propagate upstream to the MV grid. It was hypothesized from the time variation of the switching frequency that the operation of multiple UPSs in parallel at terminal LV 6 caused beat frequency oscillation. The emission from a single-phase device shows correlation which may point to conductor crosstalk at these frequencies. The presence of supraharmonic emission leads to an increase in the PE-N voltage throughout the data center. Lastly, the emissions were compared to the compatibility limits in IEC 61000-2-2 Amd 2, and it was found that the emissions are within the limits barring at 5.4 and 5.6 kHz for LV 6. The authors recommend including supraharmonics in "normal" power quality analysis. Another recommendation is to do simultaneous measurements at several locations within an installation. Lastly limits and clear guidelines to where inside an installation the said limits apply e.g., point of delivery terminal of a device and for what time-range e.g., weekly as for harmonics should be clearly defined.

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